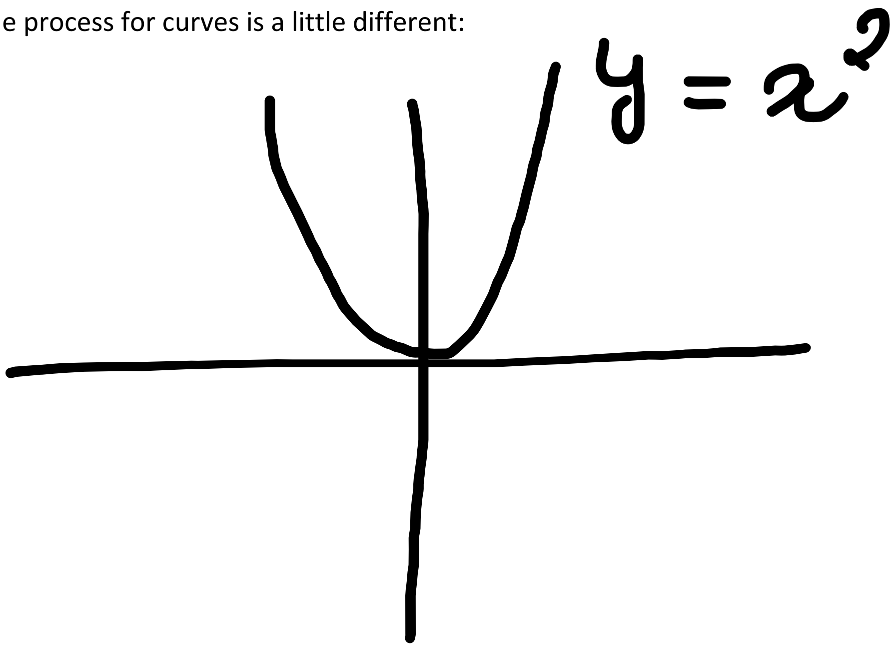
For straight-line graphs, the gradient of the line can be easily found by:

1) Rearranging the equation to the form $y = mx + c$

2) Gradient = m

However, the process for curves is a little different:



For the curve $y = x^2$ (or any curve for that matter), we can find the gradient **at a point** by drawing a tangent to that point, which gives us a straight-line graph, allowing us to use the process above.

To simplify this process, we use a derivative.

A derivative is a function that gives the gradient of the original function **at a point**.

Derivative = Gradient

Notation:

$$f'(x) = \frac{dy}{dx} = First\ derivative\ (Gradient\ function)$$

$$f''(x) = \frac{d^2y}{dx^2} = \frac{d}{dx}\left[\frac{dy}{dx}\right] = Second\ derivative\ (Derivative\ of\ gradient\ function)$$

$$\frac{d}{dx}(x^n) = n \cdot x^{n-1}$$

$$\frac{d}{dx}(k \cdot f(x)) = k\frac{d}{dx}(f(x))$$

$$\frac{d}{dx}\big(f(x) \pm g(x)\big) = \frac{d}{dx}(f(x)) \pm \frac{d}{dx}(g(x))$$

$$\frac{dy}{dx} = \frac{dy}{du} \times \frac{du}{dx}$$

Example: Differentiate $(3x^2 + 4)^{\frac{1}{3}}$

$u = 3x^2 + 4$

$\frac{du}{dx} = 6x$

$y = u^{\frac{1}{3}}$

$\frac{dy}{du} = \frac{1}{3}u^{\frac{-2}{3}}$

$\frac{dy}{dx} = \frac{1}{3}u^{\frac{-2}{3}} \times 6x = 2x(3x^2 + 4)^{\frac{-2}{3}}$

$$\frac{d}{dx}(\sin x) = \cos x$$

$$\frac{d}{dx}(\cos x) = -\sin x$$

$$\frac{d}{dx}(\tan x) = \sec^2 x$$

$$\frac{d}{dx}(e^x) = e^x$$

$$\frac{d}{dx}(\ln x) = \frac{1}{x}$$

$$\frac{d}{dx}\big(f(x) \times g(x)\big) = f(x) \times \frac{d}{dx}\big(g(x)\big) + g(x) \times \frac{d}{dx}(f(x))$$

$$\frac{d}{dx}\left(\frac{f(x)}{g(x)}\right) = \frac{g(x)f'(x) - f(x)g'(x)}{g^2(x)}$$

Derivative at a point = Gradient of tangent at that point

Gradient of tangent × Gradient of normal = -1

When the derivative (gradient) = 0, the curve has a stationary point

The second derivative is used to find maxima/minima:

$$\frac{d^2y}{dx^2} > 0 \rightarrow Minima$$

$$\frac{d^2y}{dx^2} < 0 \rightarrow Maxima$$

$\frac{d}{dt}$ represents the rate of change

Integration is the reverse process of differentiation

$$\int x^n \, dx = \frac{x^{n+1}}{n+1} + C$$

$$\int \frac{1}{x} \, dx = \ln x + C$$

$$\int \frac{1}{ax+b} \, dx = \frac{\ln|ax+b|}{a} + C$$

$$\int (ax+b)^n \, dx = \frac{(ax+b)^{n+1}}{a(n+1)} + C$$

$$\int \sin(ax+b) \, dx = \frac{-\cos(ax+b)}{a} + C$$

$$\int \cos(ax+b) \, dx = \frac{\sin(ax+b)}{a} + C$$

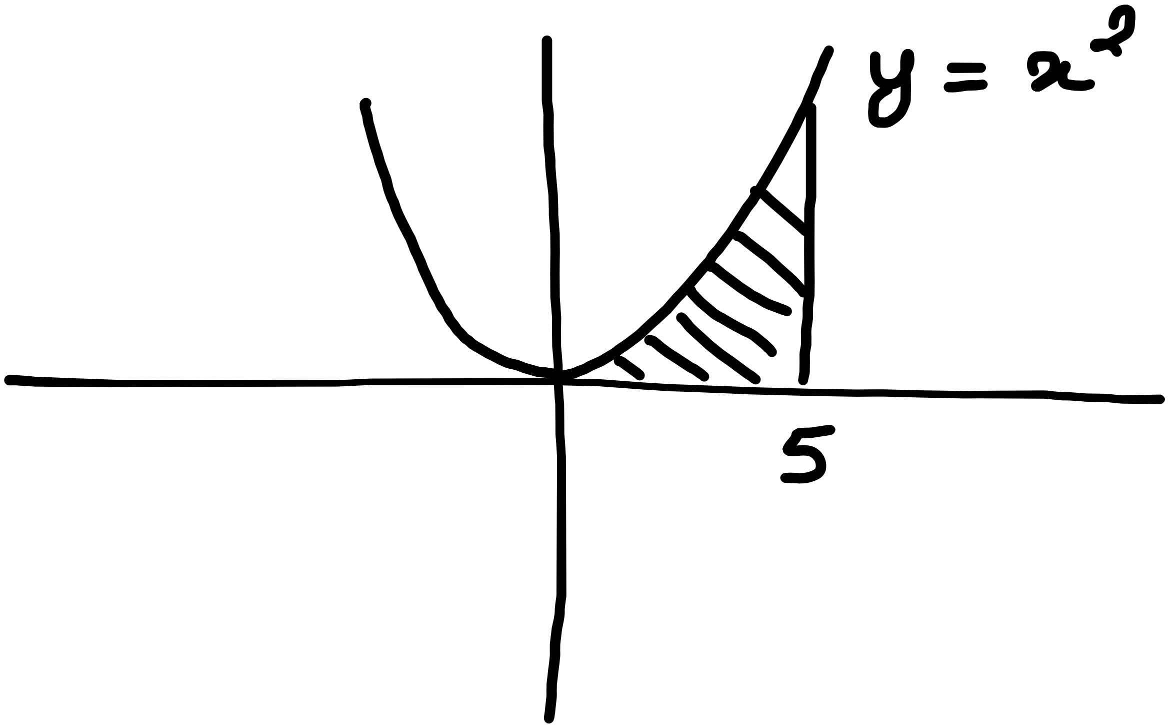
$$\int \sec^2(ax+b) \, dx = \frac{\tan(ax+b)}{a} + C$$

$$\int e^{ax+b} \, dx = \frac{e^{ax+b}}{a} + C$$

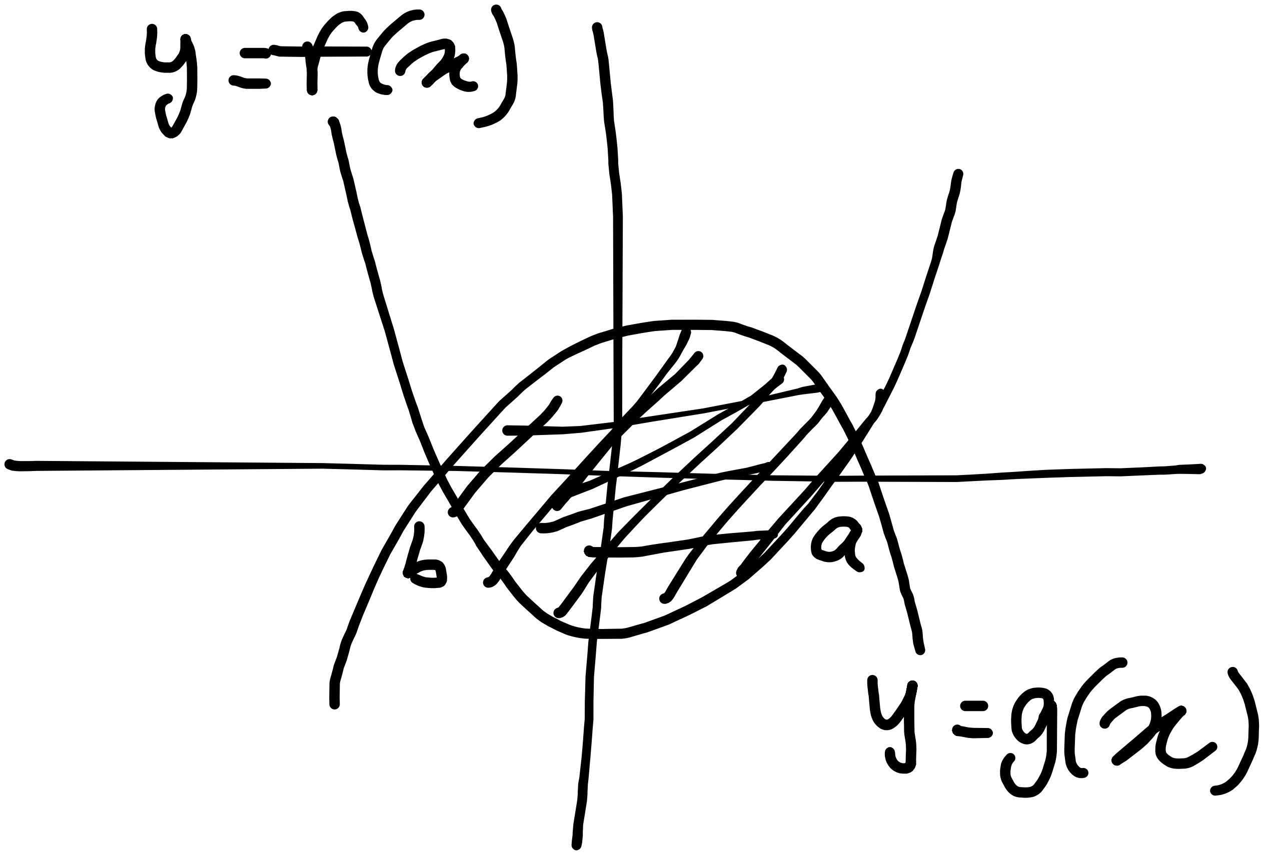
Area under a curve:

Integration can also be used to find the area under a curve

Example:



$$A = \int_0^5 x^2\,dx = \left[\frac{x^3}{3}\right]_0^5 = \frac{125}{3} - 0 = \frac{125}{3}$$



$$A = \int_b^a g(x) - f(x)\,dx$$

Kinematics:

$Displacement = s$

$Velocity = v$

$Acceleration = a$

$$v = \frac{ds}{dt}$$

$$a = \frac{dv}{dt}$$